

A SECOND LEAF, A HARDER QUESTION

Why Do Leaves Turn Red, Yellow, and Brown in Autumn?

A cost worth explaining—and a lesson in four kinds of why

One leaf is being dismantled. Three colours appear. Their optics are comparatively easy; their evolutionary histories are not. The intellectual work begins when we stop allowing one word—*because*—to conceal the difference.

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An evidence-graded essay for upper-secondary students and teachers

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Companion to *Why Are Leaves Green?*

The short answer

Autumn colour is not one phenomenon. Yellow and orange usually become conspicuous because chlorophyll disappears faster than carotenoids; those carotenoids commonly pre-date senescence, although their concentrations also change. Brown is a late-stage optical mixture produced by pigment loss, oxidation and polymerisation of phenolics, drying, and tissue breakdown—not one dedicated brown pigment. Red and purple often involve anthocyanins synthesised during senescence and stored in vacuoles.

That last fact creates a genuine evolutionary question, since synthesis uses carbon, enzymes, ATP, and reducing power. Yet the size and fitness consequence of that cost are rarely measured directly. Experiments show that anthocyanins can photoprotect senescing leaves in some species and conditions; other experiments find little such protection. Comparative studies disagree about whether red species recover nitrogen more efficiently. Anti-herbivore signalling, carbohydrate disposal, and other proposals remain in play. These mechanisms may also operate together.

The defensible conclusion is layered: pigment chemistry is well established; several present-day effects are experimentally supported but context-dependent; the historical reason or reasons that autumn redness evolved remain unsettled.

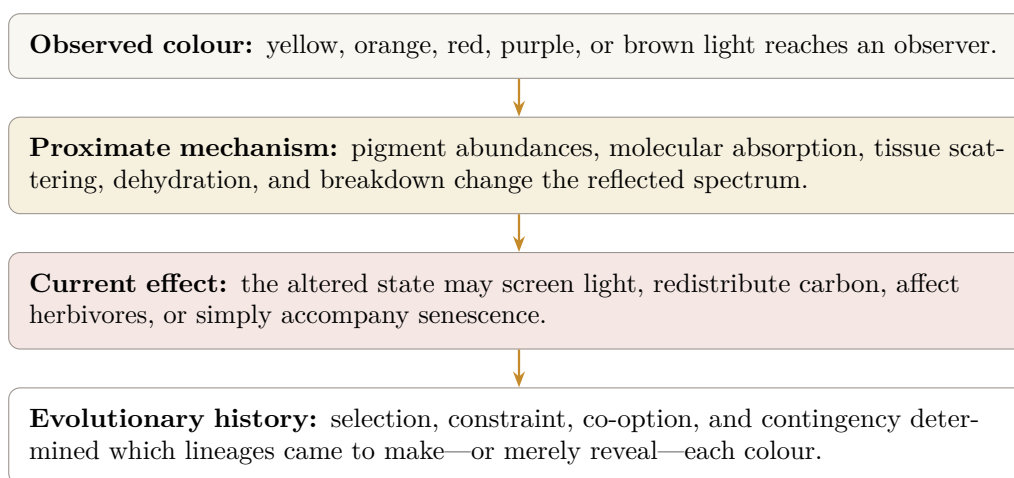
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1 Begin with the asymmetry

The original draft found the right organising idea: yellow, brown, and red carry different explanatory burdens. Its strongest image was the leaf that pays to paint itself shortly before abscission. The image survives here, with two guardrails. First, a biochemical expense is not automatically a large fitness cost. Second, identifying a current benefit does not yet identify the historical target of selection.

Autumn senescence is an orderly dismantling of the photosynthetic apparatus. Much of the nitrogen recovered from a leaf comes from abundant chloroplast proteins, while chlorophyll is catabolised through a regulated pathway that avoids the accumulation of phototoxic intermediates [??]. The colour change is therefore the visible surface of a much larger programme of remobilisation, detoxification, changing tissue optics, and eventual cell death.



Every downward arrow requires new evidence. A reflectance spectrum can establish colour. A mutant or manipulation can test a present-day effect. Comparative and phylogenetic evidence is needed for an evolutionary account. None of those methods, alone, can do all four jobs.

2 Three colours, three explanatory situations

2.1 Yellow and orange: usually revealed, never literally cost-free

Carotenoids absorb mainly blue to blue-green light and participate in light harvesting and photo-protection throughout the growing season. As chlorophyll declines during senescence, their yellow and orange contribution becomes more visible. Unmasking is a useful first approximation, but pigment pools are not frozen: carotenoids can also be retained, transformed, or degraded while chlorophyll is disappearing [??].

It is therefore too strong to say that yellow costs nothing or that no adaptive question remains. The safer claim is narrower:

In many yellow autumn leaves, conspicuous colour does not require a newly constructed seasonal display pigment.

The carotenoids were already serving functions in a green leaf. Their later visibility can be a by-product of differential pigment loss even if chlorophyll dismantling and carotenoid persistence are themselves regulated. A visible consequence need not have been selected *as a colour signal*.

2.2 Brown: a terminal state, not a single pigment

Brown is also easily over-compressed. Dead or nearly dead leaf tissue can look brown after chlorophyll and carotenoids have largely disappeared, water content has fallen, cellular order has broken down, and phenolic compounds have oxidised or polymerised into darker products. Residual pigments, tannin-rich chemistry, cell walls, air spaces, and surface structure all contribute [?].

So the claim that brown is tannin is no more complete than saying green is chlorophyll. Brown commonly names an optical outcome of late senescence and death, rather than a dedicated seasonal pigment programme. That makes a selected display function less pressing, though it does not prove that every brown trait in every species lacks ecological consequence.

2.3 Red and purple: often newly synthesised

In many deciduous species, anthocyanins accumulate during senescence in the vacuoles of epidermal or mesophyll cells. Chemically, the coloured compounds in living tissue are usually glycosylated anthocyanins; *anthocyanidins* are their aglycone cores. Light, temperature, sugar status, nutrients, development, and genotype can all affect their production [Hoch et al., 2001, Feild et al., 2001, ?].

Synthesis entails substrates, enzymes, ATP, and reducing power. Calling this a cost is reasonable at the biochemical level. Calling it a large net loss to nutrient recovery is a further empirical claim. Accumulated sugars may themselves need to be managed, and the relevant comparison is the whole-plant fitness of alternative phenotypes, not the mere existence of an anabolic pathway.

A tempting shortcut—and the repair

Shortcut: Red pigment consumes resources that the plant would otherwise recover, so red must purchase some compensating benefit.

Repair: Anthocyanin production consumes resources, but the magnitude, opportunity cost, and alternatives depend on the physiological state of the leaf. A trait may be inexpensive, multifunctional, or a way of handling surplus carbohydrate. Selection requires a fitness difference, not simply an energetic invoice.

Colour	Common proximate basis	New seasonal synthesis?	What still needs explaining?
Yellow / orange	Chlorophyll declines faster than visible carotenoids.	Often little or none for the visible colour; pigment pools remain dynamic.	Chlorophyll catabolism, carotenoid persistence, and whether visibility itself has any selected effect.
Brown	Broad pigment loss plus oxidation, polymerisation, dehydration, and tissue optics.	No single dedicated brown pathway is required.	Usually the chemistry of terminal senescence; ecological effects remain possible in particular cases.
Red / purple	Senescence-associated anthocyanin accumulation, often in vacuoles.	Commonly yes, in red-turning species.	Present functions, net costs, lineage distribution, and the historical selection pressures.

3 The hypothesis landscape: more than a duel

Two families dominate the literature, photoprotection and coevolutionary signalling. They are prominent rather than exhaustive. Reviews have also discussed antioxidant effects, carbohydrate disposal, osmotic or drought-related roles, leaf warming, camouflage or anti-camouflage, fruit signalling, biochemical constraint, and nonadaptive by-products [???].

3.1 Photoprotection and nutrient resorption

The photoprotection hypothesis proposes that anthocyanins screen excess visible light or help manage oxidative stress when cool temperatures and dismantling photosystems make absorbed energy difficult to use. Protection could keep leaf cells functional long enough to remobilise more nutrients.

Several experiments support parts of that causal chain. In red-osier dogwood, anthocyanic tissues attenuated light reaching chloroplasts and red-senescent leaves recovered photosystem function better after strong-light treatment than yellow tissues [Feild et al., 2001]. Work with anthocyanic and anthocyanin-deficient material in woody plants linked pigment presence to lower photoinhibition and, under the tested conditions, more complete nutrient resorption [Hoch et al., 2003].

These are important functional results. Their scope matters. They show what anthocyanins *can do* in particular systems; they do not establish that every autumn-red lineage evolved for that reason.

3.2 Coevolution, signalling, and herbivores

The coevolutionary hypothesis proposes that autumn colour affects insects that choose hosts at the end of the growing season. In one version, bright colour honestly advertises defensive commitment, discouraging aphids from colonising a tree [Archetti, 2000, Hamilton and Brown, 2001]. Other animal-facing ideas involve concealment from animals with limited red sensitivity, or making herbivores more conspicuous to their predators.

The evidence is mixed and the causal target is slippery. Redness around damaged branches in Norway maple, for example, complicates a simple whole-tree signal of vigour [Sinkkonen, 2008]. Critics have also asked whether insects perceive the relevant colours as assumed and whether colour predicts defence after phylogeny, condition, and correlated chemistry are controlled [Schaefer and Wilkinson, 2004].

A 2025 phylogenetic meta-analysis found that nongreen leaves, taken across developmental stages and colour systems, tended to be better defended, less nutritious, and less herbivorised than green leaves [?]. That is relevant new evidence. It does not uniquely validate autumn-red signalling: the synthesis combined red, yellow, purple, white, and variegated leaves, and only a small subset concerned autumn colour. Association among colour, chemistry, and herbivory also leaves open whether colour is a signal, a camouflage device, a correlated marker, or part of a broader defence syndrome.

3.3 Carbon balance and multifunctionality

A senescing leaf can accumulate sugars as export, metabolism, temperature, and photosynthetic capacity change. Anthocyanin synthesis may provide a temporary carbon sink or participate in redox and osmotic regulation. A 2023 study of different coloured regions within Norway maple leaves found no significant photoinhibition difference between red and yellow senescing tissue and proposed carbohydrate disposal as an alternative interpretation [?].

That result does not disprove photoprotection everywhere. It shows why the propositions that anthocyanins photoprotect and that autumn red evolved for photoprotection must not be treated as universal synonyms. Different lineages, tissues, stages, and environments can yield different functional balances.

Provenance check

Marco Archetti originated the *coevolutionary* proposal; his later comparative work challenged the rival photoprotection account. The story is methodological: experiments, cross-species

comparisons, reanalyses, and within-leaf studies probe different scales and reach different conclusions.

4 A disagreement worth teaching

Pena-Novas and Archetti [2021] compared mature and senescent leaf nitrogen across 55 tree species. They found no association between anthocyanin presence and nitrogen-resorption efficiency; yellow species showed more efficient resorption in their analysis. They interpreted this as evidence that chlorophyll loss, rather than anthocyanin production, better tracked nitrogen translocation across those species.

Joly [2022] reanalysed the same data using a phylogenetic comparative method and found greater resorption in species with red or yellow autumn leaves than in species remaining green. ? replied that red and yellow species still did not differ significantly, leaving anthocyanin's additional contribution unresolved.

This exchange is more informative than a clean winner:

- the same dataset can yield different contrasts under different models;
- coloured versus green tests a different claim from red versus yellow;
- species categories conceal variation in pigment amount, timing, leaf mass loss, and environment;
- correlation across species cannot by itself identify the causal work of a pigment;
- phylogenetic correction improves an inference, but does not supply missing measurements.

The lesson is not that statistics can say anything. It is that a biological hypothesis has to specify its scale and its predicted comparison. A leaf-level protection effect, a species-level association, and an evolutionary origin are related propositions, not interchangeable ones.

5 What is settled, and what is not?

Claim	Evidence grade	What the evidence warrants
Autumn yellow commonly becomes visible as chlorophyll declines relative to carotenoids.	Established mechanism	Pigment chemistry and senescence studies support this; unmasking remains a simplification because carotenoids also change [??].
Brown is simply tannin left behind.	Overstated	Late brown can include residual pigments, oxidised or polymerised phenolics, dehydration, cell death, and altered scattering. There is no single universal brown pigment.
Many red-turning leaves synthesise anthocyanins during senescence.	Established mechanism	Biochemistry, time courses, and tissue measurements agree [Hoch et al., 2001, Feild et al., 2001].
Anthocyanins can protect senescing photosystems under excess light.	Strong but bounded	Supported in several species and treatments [Feild et al., 2001, Hoch et al., 2003]; weak or absent in other systems, including a within-leaf Norway maple study [?].
Red leaves therefore recover more nitrogen across species.	Live dispute	Comparative analyses of the same 55-species dataset disagree about some contrasts; red and yellow remain difficult to separate [Pena-Novas and Archetti, 2021, Joly, 2022, ?].
Autumn colour is an honest warning to aphids.	Contested hypothesis	There are suggestive behavioural and comparative patterns, alongside failed or ambiguous predictions and correlated traits [Schaefer and Wilkinson, 2004, Sinkkonen, 2008, ?].
Photoprotection and coevolution are mutually exclusive.	False dichotomy	A pigment may have physiological and ecological consequences simultaneously; nutrient status can connect them [?].
Redness was historically selected for one known reason.	Open inference	Current evidence supports several functions in some contexts, without uniquely reconstructing the original or continuing selection pressure.

6 A teleology caught in the act

When Feild et al. [2001] asked Why leaves turn red in autumn, Harry Beevers objected that the title encouraged a teleological reading. His published letter argued that pigment production supplies the proximate answer and warned against assuming in advance that redness must benefit the plant [Beevers, 2002]. His criticism usefully detects a logical risk, though his proposed remedy is too restrictive. Biology is entitled to ask about function and evolutionary history as well as mechanism. The safeguard is to label the question and demand the appropriate evidence.

Biologists have long separated proximate and evolutionary causes [?]; Tinbergen distinguished mechanism, development, function, and phylogeny [?]. Dennett later explored how natural selection can generate design-like reasons without a foresighted designer [?]. The following fourfold adapts those traditions to one autumn leaf.

Question	Kind of answer	Autumn-red answer, with its proper confidence
1. How is it so?	Mechanism	Senescence-associated regulation activates flavonoid biosynthesis; anthocyanins accumulate in vacuoles and alter spectral reflectance and transmittance. Well established.
2. How does it become so?	Development	Light, temperature, sugars, nutrient state, hormones, and genotype affect when and where a leaf reddens. Species and even regions within one leaf can differ. Broadly secure; details variable.
3. What does it do now?	Current causal role	Depending on the system, pigment may screen light, alter redox or carbon balance, and affect animals. Effects are experimentally testable and context-dependent. Several supported roles; no universal profile.
4. Why did this lineage retain it?	Evolutionary history / selected effect	Photoprotection, nutrient economics, herbivore interactions, multifunctionality, constraint, and co-option remain candidate parts of the history. Open and probably plural.

There is a further question hidden in ordinary speech: **Who meant it?** Natural selection can preserve a variant because of its consequences without any organism anticipating an autumn display. That is a reason in the historical and functional senses. It is not conscious purpose.

7 How to turn the stories into harder tests

7.1 Manipulate pigment within a shared background

Mutants, closely matched colour morphs, or red and yellow regions within the same leaf can reduce confounding by species history. Researchers can quantify incident and transmitted light, photosystem II damage and repair, oxidative markers, sugar export, nitrogen content, and final abscission. The strongest design measures several links in the proposed chain rather than treating redness itself as proof of protection.

7.2 Measure cost instead of merely naming it

The claim that a leaf pays for anthocyanin should generate estimates: carbon incorporated into pigment, enzyme and transport investment, ATP and reductant demand, the fate of alternative sugars, and the effect on whole-plant survival or reproduction. A pathway can require energy while still solving a disposal problem; a gross biosynthetic cost and a net fitness cost are different quantities.

7.3 Separate the comparisons

A useful study distinguishes at least three contrasts:

1. red tissue versus matched yellow tissue;
2. red or yellow tissue versus tissue that remains green;
3. red-producing lineages versus related non-red lineages across environments.

These contrasts answer different questions. The first isolates anthocyanin more closely. The second may mainly track chlorophyll loss. The third addresses evolutionary distribution but introduces phylogeny and ecological covariation.

7.4 Make rival explanations predict different observations

Photoprotection predicts benefits that strengthen under damaging light-temperature combinations and should be measurable in photosystem performance or successful nutrient export. A signal hypothesis predicts receiver-specific perception, behaviour, and fitness consequences, plus a reliable relation between colour and defence. A carbohydrate-sink hypothesis predicts redness where export is constrained or sugar accumulates, even when photoprotection is weak. Multifunctionality predicts interactions rather than one universal winner.

Question worth keeping open

What result would make you lower your confidence? If every outcome is declared consistent with a hypothesis, the hypothesis is decorating the evidence rather than risking contact with it.

8 What school science should do with an unfinished answer

School knowledge has to be compressed. The educational problem begins when a compression silently changes its type:

Useful first mechanism: Chlorophyll loss reveals carotenoids; many red leaves also synthesise anthocyanins.



Unmarked generalisation: Yellow is free, brown is tannin, and red is costly.



Origin story presented as fact: Trees make red pigment to protect nitrogen recovery—or to warn aphids away.

The first sentence is a sound entry point. The second turns useful tendencies into universals. The third converts a candidate adaptive explanation into settled history. Students can learn to see those upgrades without concluding that scientific knowledge is arbitrary.

8.1 A six-line provenance card

Claim	State the smallest exact proposition. Do not combine mechanism, effect, and origin in one because.
Kind of why	Observation, mechanism, development, current role, evolutionary history, or conscious purpose?
Evidence	Spectrum, pigment assay, mutant, manipulation, time series, behaviour test, comparative model, or review?
Scope	Which species, tissue, light level, temperature, life stage, and comparison?
Confidence	Established; supported but bounded; actively disputed; plausible; or speculative?
Source chain	Classroom statement → synthesis or review → original data. Does each source actually make the same claim?

8.2 A classroom investigation with honest limits

Students can sample leaves from sun-exposed and shaded parts of one tree, photograph them with a colour standard, and compare colour with position and chlorophyll estimates. That design can

test whether redness is *associated* with exposure within the sampled tree. It cannot, on its own, establish photoprotection, nitrogen resorption, or evolutionary origin. Leaf age, branch health, orientation, and temperature are plausible confounders and should be recorded.

A stronger desk-based exercise uses published data. Give students the 55-species colour and nitrogen dataset discussed by [Pena-Novas and Archetti \[2021\]](#) and [Joly \[2022\]](#); ask them to state the red–yellow and coloured–green predictions separately. Then compare how analytical choices change the inference. The disagreement becomes a lesson in model specification rather than a slogan about uncertainty.

9 A reading route for students

1. **Learn the dismantling.** Read the overview of chlorophyll catabolism by ? or the shorter synthesis by ?. Ask which components of a chloroplast are remobilised and which are detoxified.
2. **See the classic photoprotection case.** Read the abstract, figures, and discussion of [Feild et al. \[2001\]](#), followed by [Hoch et al. \[2003\]](#). List the measured links in the proposed chain from pigment to protection to nutrient recovery.
3. **Watch why become contested.** Read Beevers’s one-page letter [[Beevers, 2002](#)]. Decide where his warning is valid and where it makes biological explanation too narrow.
4. **Meet the evolutionary alternatives.** Use ? for the hypothesis map and ? for the interdisciplinary synthesis. Notice how many proposals disappear when a school account says there are only two.
5. **Read a data disagreement, not a summary of it.** Compare [Pena-Novas and Archetti \[2021\]](#), [Joly \[2022\]](#), and the reply by ?. Write down exactly which group contrast changes and which does not.
6. **Test generality.** Compare the positive functional evidence with the within-leaf Norway maple result of ?. Ask whether a negative result challenges a universal claim, a species-specific claim, or both.
7. **Enter the current frontier.** Read the complementary-hypotheses argument of ?, the broad review by ?, and the 2025 meta-analysis by ?. Check how much of each paper is specifically about autumn-red leaves.
8. **Sort the whys.** Reclassify every because you have written as mechanism, development, present effect, historical selection, or intention. Where a sentence does two jobs, split it.

10 Coda: the leaf that pays—perhaps

A yellow leaf often reveals pigments it already had. A brown leaf displays the chemistry and optics of tissue nearing death. A red leaf commonly activates a biosynthetic programme while the organ is being dismantled. That contrast makes red a sharper question, though the metaphor of payment must remain an invitation to measure rather than a conclusion smuggled into the premise.

The research record supplies several disciplined possibilities. Anthocyanins can screen light and sometimes preserve photosystem function. They can coincide with nutrient economy, sugar accumulation, stress, and animal interactions. Their effects differ across species and conditions. Cross-species analyses disagree about the contrast that matters. A physiological function and an ecological signal may be complementary, and either may be a later use of a pathway inherited for other reasons.

If a future synthesis reconstructs a causal history containing constraints, accidents, co-options, and episodes of selection, that history will be a perfectly good reason why. It will not require a tree to foresee October. Meaning and conscious purpose belong to another register; biological function can emerge before there are reasoners capable of naming it.

The most valuable classroom answer is therefore neither a confident fable nor a shrug. It is a layered judgement:

We know how many autumn colours are produced. We know that some proposed functions occur in some systems. We do not yet know one universal evolutionary reason for autumn redness, and the live disagreement tells us exactly what further evidence would have to discriminate.

In an age of easy answers, provenance is part of the content. A scientific statement becomes stronger when a learner can see its scale, method, uncertainty, and route back to evidence. The changing leaf offers a vivid place to practise that form of discernment.

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